

Keon E

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SUMMARY OF QUALIFICATIONS

- Built and trained neural networks from scratch in C++, reaching **98.02%** validation accuracy on MNIST
- Programming in C++ (strongest) and Python, with experience in Git-based team environments
- Hands-on data preprocessing and model evaluation from writing my own training pipeline
- Learned a production cluster's workflow on my own through its documentation as a WATcloud contributor
- Communication and leadership from supervising pool attendants for the City of Ottawa

EDUCATION

University of Waterloo

Waterloo, ON

Honours Mathematics, Co-operative Program

Aug. 2026 – Expected 2031

Relevant Coursework: Designing Functional Programs (CS 135), Algebra (Advanced) (MATH 145), Calculus 1 (Advanced) (MATH 147)

Euclid Mathematics Contest: Distinction, top 25%

PROJECTS

Tensor Autograd Engine | C++ | [GitHub](#)

2026

- Built a scalar autograd engine and extended it to N-dimensional tensors with broadcasting, matrix multiplication, convolution, and pooling
- Implemented backpropagation, tensor memory layout, and a full training pipeline with no external ML libraries
- Trained and evaluated an MLP to **97.34%** and a CNN to **98.02%** validation accuracy on MNIST
- Fixed a memory leak caused by a reference cycle between tensors and their backward functions

EXPERIENCE

WATcloud

Waterloo, ON

Team Member, Student-Run Compute Cluster

Sep. 2026 – Present

- Added kernel version reporting to the cluster's machine cards, completing an issue a previous member had started; pull request reviewed and merged
- Wrote documentation for using the cluster VPN from the command line; currently working on an issue involving Terraform

City of Ottawa

Ottawa, ON

Wading Pool Supervisor

Summer 2025

- Promoted to Supervisor after two summers as Pool Attendant
- Supervised attendant staff and oversaw daily pool operations, enforcing safety protocols and City guidelines

Pool Attendant

Summers 2023 – 2024

- Monitored swimmer safety and supported site staff in daily wading pool operations

TECHNICAL SKILLS

Languages: C++, C, Python, Racket

Machine Learning: Backpropagation, neural networks (MLP, CNN), data preprocessing, model evaluation

Tools: Git, SSH, SLURM, g++

EXTRACURRICULAR ACTIVITIES

Lisgar Collegiate Institute: Varsity Basketball (2025 – 2026), Track and Field (2023 – 2026), Symphonic Winds, Bb Clarinet (2022 – 2026)